Science

Class 4 (English Medium)

Solved Exercises — Sindh Textbook Board

10 lessons — 233 questions

Our Body

Review Questions: 1. Checking for Vocabulary — Unscramble the Words

Question: 1. Unscramble the word: EEY

Answer: EYE. The jumbled letters EEY make the word EYE. The eye is the part of our body with which we see the things around us.

Question: 1. Unscramble the word: ETRAH

Answer: HEART. The jumbled letters ETRAH make the word HEART. The heart pumps blood to all parts of the body.

Question: 1. Unscramble the word: GNUL

Answer: LUNG. The jumbled letters GNUL make the word LUNG. The lungs are our breathing organs, and with their help we breathe in and out.

Question: 1. Unscramble the word: OEBN

Answer: BONE. The jumbled letters OEBN make the word BONE. Bones support our body and give it shape.

Question: 1. Unscramble the word: EEHTT

Answer: TEETH. The jumbled letters EEHTT make the word TEETH. Teeth bite, chew, crush and grind the food that we eat.

Question: 1. Unscramble the word: UMLSCE

Answer: MUSCLE. The jumbled letters UMLSCE make the word MUSCLE. Muscles work together with the bones and help the body to move.

Question: 1. Unscramble the word: OETNSKEL

Answer: SKELETON. The jumbled letters OETNSKEL make the word SKELETON. The skeleton is the framework of all the bones, which gives shape and support to the body.

2. Circle the Best Response

2. (i) Digestion begins in one of the following parts of the body:

(a) In the stomach

(b) In the lungs

✓ (c) In the mouth

(d) In the heart

2. (ii) The largest part of the body:

(a) Heart

✓ (b) Skin

(c) Brain

(d) Stomach

2. (iii) The part of the body that protects the internal body parts:

(a) Muscles

(b) Brain

✓ (c) Skin

(d) Ear

2. (iv) The part of the body that controls the overall body movements:

(a) Heart

✓ (b) Brain

(c) Muscles

(d) Bones

2. (v) The part of the body that gives you shape is:

(a) Stomach

✓ (b) Skeleton

(c) Muscles

(d) Skin

3. Write the Key Functions of the Following Body Parts

Question: 3. Write the key function of the Eye.

Answer: The eye is used for seeing. With the help of our eyes we can see the things around us, their colours and their shapes.

Question: 3. Write the key function of the Teeth.

Answer: Teeth bite, chew, crush and grind the food we eat. In this way the food breaks into small pieces and is digested easily in the stomach.

Question: 3. Write the key function of the Nose.

Answer: The nose is used for breathing and for smelling. Through the nose air reaches the lungs, and we can recognise different smells.

Question: 3. Write the key function of the Muscles.

Answer: Muscles move the bones. When muscles contract and relax, the bones move, and so we can walk, run and lift things.

Question: 3. Write the key function of the Skeleton.

Answer: The skeleton gives shape and support to the body. It also protects the delicate internal parts such as the brain, the heart and the lungs.

Question: 3. Write the key function of the Heart.

Answer: The heart pumps blood to all parts of the body. In this way food substances and oxygen reach every part of the body.

Question: 3. Write the key function of the Lungs.

Answer: The lungs are the main organs of breathing. They take oxygen from the air into the blood and remove the waste gas from the body.

Question: 3. Write the key function of the Stomach.

Answer: The stomach stores food and digests it. The juices of the stomach make the food soft and break it into small substances so that the body can use it.

4. Answer the Following Questions

Question: 4. (a) Name disorders of teeth, stomach, heart and lungs and

suggest some ways to prevent them.

Answer: Disorders of the teeth are cavities and toothache; to prevent them we should brush our teeth twice a day and eat fewer sweet things. Disorders of the stomach are indigestion and stomach ache; we should eat clean and fresh food on time. Disorders of the heart are fast heartbeat and heart diseases; we should exercise daily and eat less oily food. Disorders of the lungs are cough and difficulty in breathing; we should breathe in clean air and stay away from tobacco smoke.

Question: 4. (b) Suggest some precautionary measures to athletes and motor cyclists to safeguard their bones while engaging in sports and traveling.

Answer: Athletes should warm up their bodies before playing, wear protective things such as knee pads and elbow pads, and follow the rules of the game. Motor cyclists should always wear a helmet, drive slowly and carefully, and follow the traffic rules. With these precautions the bones can be saved from fractures and injuries.

5. Place a Yes for Correct Statement and No for a Wrong Statement

✓ 5. (i) Human body is composed of different parts.

✓ 5. (ii) Brain is protected by skull.

✗ 5. (iii) Cavities can develop in teeth if you brush them properly.

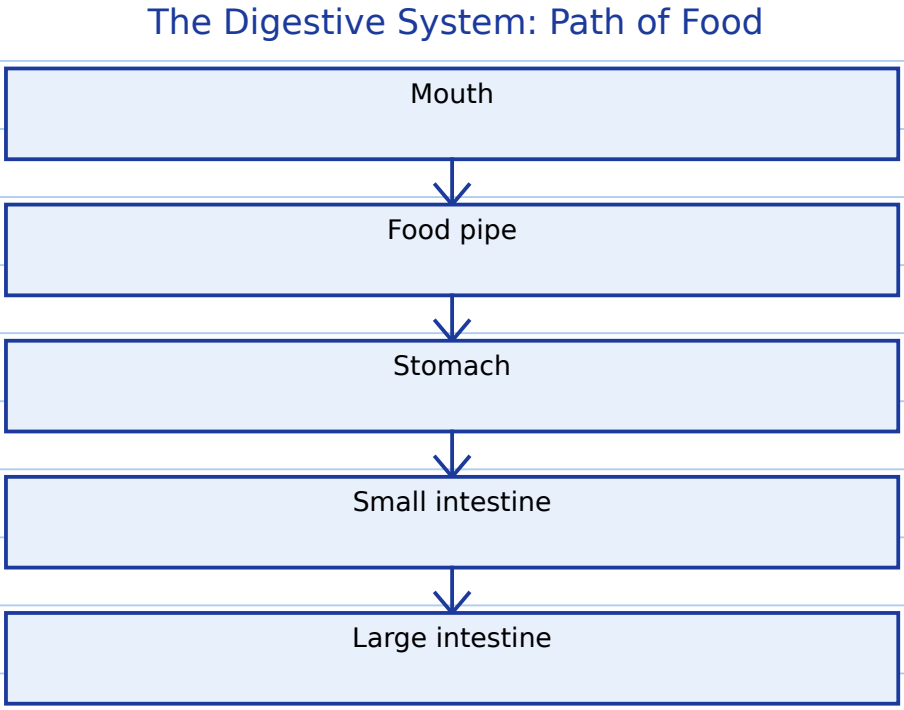
✗ 5. (iv) Ears are the organs of sight.

✓ 5. (v) We have 27 bones in our one hand.

Project: Model Making — Construct a Model of Body Parts that Digest Food

Question: Project: Using chart paper, a box, a rubber tubing, polythene bags, a small plastic bottle, a used light bulb and a string, how will you construct a model of the digestive system?

Answer: This is a practical activity; do it like this: first investigate and find diagrams of the human digestive system and decide which thing can be used for which part. Use the tubing as the food pipe, the bottle or bag as the stomach and a long tubing as the intestines, and join all the parts in order. Place the model inside a large box and demonstrate how the intestine passes through the diaphragm and how food is digested.



Characteristics and Needs of Living Things

Review Questions: 1. Circle T for True and F for False


1. (i) Non-living things can grow.


1. (ii) Living things need Air.


1. (iii) In the life cycle of plant, the leaves grow first.


1. (iv) A baby elephant inherits all qualities from the mother elephant.

2. Compare the Two Parrots: Write Their Similarities and Differences

Question: 2. Compare the two parrots given in the book (the green parrot and the white parrot). How are they similar and different? Write their similarities and differences.

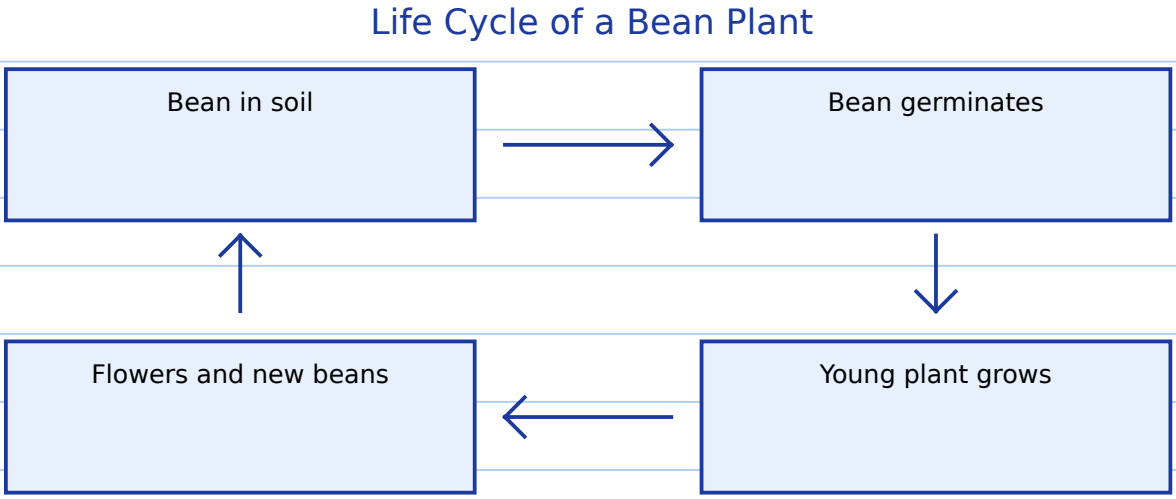
Answer: Similarities: both are parrots, both have feathers, a beak and two legs, both can fly, and both eat seeds and fruits. Differences: the first parrot is green and has a red beak, while the second parrot is white and has a yellow crest on its head. The green parrot lives on trees; the white parrot can also be seen walking on the ground.

3. Use the Words to Label the Life Cycle of the Plant

Question: 3. Using the given words (Life Cycle, Grow, Growth, Bean, Soil, Germinate, Stem, Roots, Flower, Shoot), write the life cycle of a bean plant in order.

Answer: The cycle goes in this order: first the bean (seed) is sown in the soil. With water and warmth the bean germinates; the roots grow down into the soil and the stem and shoot grow upwards. Then the plant grows and its growth continues with new shoots and leaves. At last flowers appear on the plant, new beans are formed from them, and the life cycle

starts again.



Activity: Observing a Sown Seed — Write the Stages at Different Times After Sowing

Question: Sow a seed and write what stages are seen after 3 days of sowing, after 7 days, after two weeks and after one month.

Answer: After 3 days: the seed swells and a tiny white root begins to come out of it. After 7 days: the root goes deeper into the soil and a small stem comes out of the ground. After two weeks: the stem becomes bigger and the first green leaves appear on it. After one month: the plant has grown quite big, with roots, a stem and many leaves.

4. Group Activity: Sequencing the Cards of a Germinating Seed

Question: 4. Arrange the cards of a sprouting bean in the correct order.

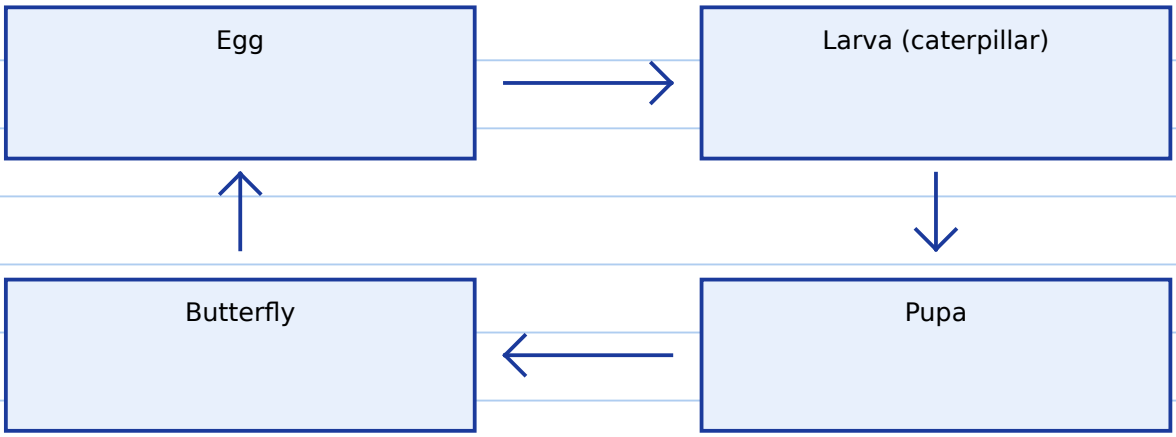
Answer: The correct order is this: first there is the dry seed (bean). Then the seed absorbs water, swells and its coat bursts. After that the root comes out of the seed and grows downwards. Then the stem grows upwards and the seed comes out of the ground. At last a small plant with roots, stem and leaves is formed.

Project Activity: Investigating the Life Cycle of Butterfly

Question: Project: Using an empty glass or plastic bottle with small holes in its lid, a larva and fresh leaves, observe the life cycle of a butterfly. Why are the holes in the lid necessary, and why should fresh leaves be changed daily?

Answer: This is a practical activity; do it like this: place the larva and fresh leaves in the bottle, cover it with the lid having holes and keep it in an airy place. The holes in the lid are necessary so that the larva gets fresh air to breathe, otherwise it would die. Fresh leaves should be changed daily so that the larva keeps getting fresh leaves to eat. In some days the larva becomes a pupa, and then a beautiful butterfly comes out of it.

Life Cycle of a Butterfly



Food and Health

Review Questions: 1. Circle T for True and F for False Statements

1. (a) A balanced diet includes food from three food groups. ✗
1. (b) A ruptured skin requires protein rich food for its repair. ✓
1. (c) In winter, half portion of our daily diet should comprise Fat. ✗
1. (d) Vitamins do not provide energy but are important for health. ✓
1. (e) A Food Pyramid Chart helps one in choose suitable food daily. ✓

2. Give Answers for the Following Questions

Question: 2. (i) What are the main sources of protein?

Answer: The main sources of protein are meat, milk, eggs, fish, pulses and peas. These foods are very necessary for the growth of the body and for repairing its worn-out parts.

Question: 2. (ii) What is similar and different in Fats and Carbohydrates?

Answer: Similarities: both give us energy and both are a part of our daily diet. Differences: carbohydrates give us quick energy and are obtained from grains, sugar and bread, while fats store energy and give warmth to the body and are obtained from oil, butter and ghee.

Question: 2. (iii) What is similar and different in Vitamins and Minerals?

Answer: Similarities: both do not give us energy, but both are necessary for health and both are needed in small amounts. Differences: vitamins are obtained from fruits and vegetables and protect the body from diseases, while minerals such as salt and calcium are obtained from milk, meat and vegetables and help in the growth of bones and teeth.

Question: 2. (iv) A food pyramid helps one to prepare a balanced diet plan. Give three reasons.

Answer: First reason: the pyramid tells us how much food of each group should be eaten daily. Second reason: with its help we can include food from all the groups in our diet, and no group is left out. Third reason: with the help of the pyramid we can avoid eating too much or too little and can prepare a healthy and balanced diet.

Question: 2. (v) Give three effects of eating unbalanced diet on health.

Answer: First effect: the body becomes weak and we get tired quickly. Second effect: the growth of children stops and they fall ill easily. Third effect: the shortage of any one substance can cause special diseases, such as weak eyesight due to shortage of vitamins and shortage of blood due to shortage of iron.

Project: Preparing a Balanced Diet Plan for My Family

Question: Project: Prepare a balanced diet plan for three days for your family, in which each day includes food from at least three food groups and the three days together include food from every food group.

Answer: A sample plan: Day 1: milk and roti in breakfast, rice and pulses in lunch, vegetable and roti at night, with a banana. Day 2: egg and milk in breakfast, meat and roti in lunch, pulses and vegetable at night, with an apple. Day 3: porridge and milk in breakfast, fish and rice in lunch, peas and roti at night, with an orange. Add clean water and milk as drinks every day. This plan includes all the groups: grains, fruits, vegetables, dairy, and meat/pulses.

Question: Project Activity Question 1: What sources of food did you include in the plan?

Answer: In the plan we included both plant sources and animal sources of food. From plants we took grains, pulses, vegetables and fruits, and from animals we took milk, eggs, meat and fish, so that the diet becomes

complete and balanced.

Question: Project Activity Question 2: Which source do you eat the most?

Answer: We eat plant food the most, especially grains such as rice and wheat roti, because they are included in every meal and give us energy. We also eat pulses and vegetables daily.

Question: Project Activity Question 3: Which source do you eat the least?

Answer: We eat fish the least, because it is not available daily in our area and it is also rather expensive. We eat meat only once or twice a week.

Living Things and their Environment

Review Questions: 1. Circle T for True and F for False Statements

✓ 1. (a) Land is not a living component of the environment.

✓ 1. (b) Human being is a living component of the environment.

✗ 1. (c) Grasslands have many trees.

✓ 1. (d) Wetland is a water environment.

✗ 1. (e) Herbivores eat plant and animals.

✓ 1. (f) Animals that eat herbivores are secondary consumers.

✗ 1. (g) Animals that eat carnivores are secondary consumers.

✓ 1. (h) All Food Chains start with the Sun.

2. Circle the Best Answer

2. (i) What is not included in the living components of the land environment?

(a) Flowering Plants

(b) Frogs

✓ (c) Iron Rock

2. (ii) Which is not a type of environment?

(a) Garden

✓ (b) Storeroom

(c) Pond

2. (iii) Which of these are not found in the food web?

✓ (a) Air

(b) Sun

(c) Human

2. (iv) If plants have needle like leaves to which environment would they

belong?

✓ (a) Forest

(b) Desert

(c) Grassland

2. (v) What is the key role of decomposers?

✓ (a) Break down dead animals

(b) Break down living plants

(c) Provide food for animals

2. (vi) In which group would you place human beings?

(a) Carnivores

✓ (b) Omnivores

(c) Herbivores

2. (vii) What is the correct order of the levels in a food chain?

(a) Consumer - Producer - Decomposer

(b) Decomposer - Consumer - Producer

✓ (c) Producer - Consumer - Decomposer

2. (viii) Which one of the following animal is a decomposer?

(a) Wolf

✓ (b) Mould

(c) Bird

2. (ix) Use the food chain Sun-Plant-Frogs-Snakes-Hawks: Which living thing would decrease, if the number of frogs increased?

(a) Hawks

(b) Snakes

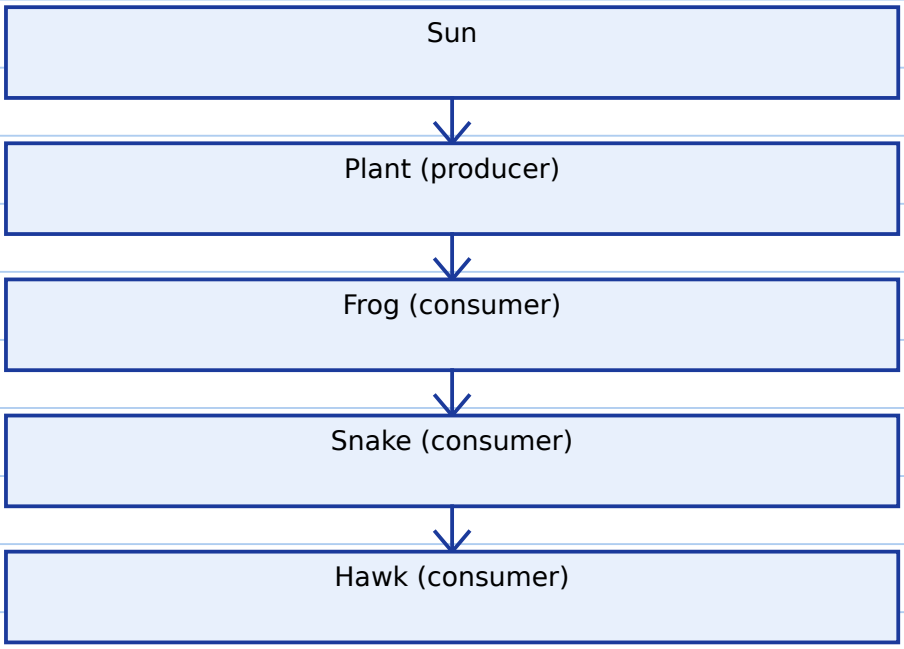
✓ (c) Plants

Question: 2. (x) Which best describes the frog? (a) Decomposer (b) Consumer (c) Producer

Answer: (b) Consumer. A frog cannot make its own food; it eats other living

things, so it is a consumer.

A Simple Food Chain



3. Give Reasons of the Following

Question: 3. Give a reason: Decomposers are an important part of the food chain.

Answer: Decomposers break down dead plants and animals into small substances, which mix into the soil. This makes the soil fertile and plants get the substances they need for growth. If there were no decomposers, dead bodies would keep collecting and the land would fill with waste.

Question: 3. Give a reason: Plants and animals are important for each other.

Answer: Plants give food and oxygen to animals and also give a home to many animals. On the other hand, animals give carbon dioxide to plants and help in spreading their seeds and in the pollination of flowers. That is why both cannot live without each other.

Question: 3. Give a reason: The environment has many living and non-living components.

Answer: Everything around us is a part of the environment. Living components include plants, animals and human beings, and non-living components include air, water, soil, rocks and sunlight. All living things depend on these non-living components for their life, so the environment is made of both kinds of components.

Project: Making Different Environments — Answer the Activity Questions

Question: Project Activity Question 1: What is common in all environments?

Answer: All environments have both living and non-living components. In every environment there are non-living components like air, water or soil and living components like plants and animals, and all living things depend on one another for life.

Question: Project Activity Question 2: What is different in all environments?

Answer: The weather, the amount of water and the kind of land are different in every environment. That is why the plants and animals living in each environment are also different, and their characteristics are according to their environment, such as the camel and cactus in the desert and fishes in water.

Question: Project Activity Question 3: What changes have we humans introduced in these environments?

Answer: Humans have cut trees and finished forests, made air and water dirty with the smoke of factories and vehicles, and reduced the homes of animals by building cities and roads. Due to these changes many animals and plants have come in danger. We should plant trees, reduce waste and protect the environment.

Matter and its States

Review Questions: 1. Circle T for True and F for False Statements


1. (i) Solids do not take the shape of the container they are kept in.


1. (ii) Gases do not change volume when transferred from a small ball to a large ball.


1. (iii) Liquids take the shape of a container they are placed in.


1. (iv) Solids that dissolve in water can be separated through filtration.


1. (v) Tea leaf can be separated from tea by filtration.

2. Explain with Examples

Question: 2. (A) Explain with examples: All forms of Matter (solid, liquid and gases) have mass and occupy space.

Answer: Anything that has mass and occupies space is matter. Example: a book is a solid; it has mass and it occupies space on the table. Milk is a liquid; it also has mass and occupies space in the glass. The air inside a balloon is a gas; it also has a little mass and occupies all the space of the balloon.

Question: 2. (B) Explain with examples: Ice changes to water on heating.

Answer: Ice is the solid form of water. When we heat ice, it melts and changes into water, because with heat its particles become loose and start flowing. For example, an ice lolly and ice cream melt when kept in a warm place.

Question: 2. (C) Explain with examples: Sugar dissolves in water and plastic pieces float on water.

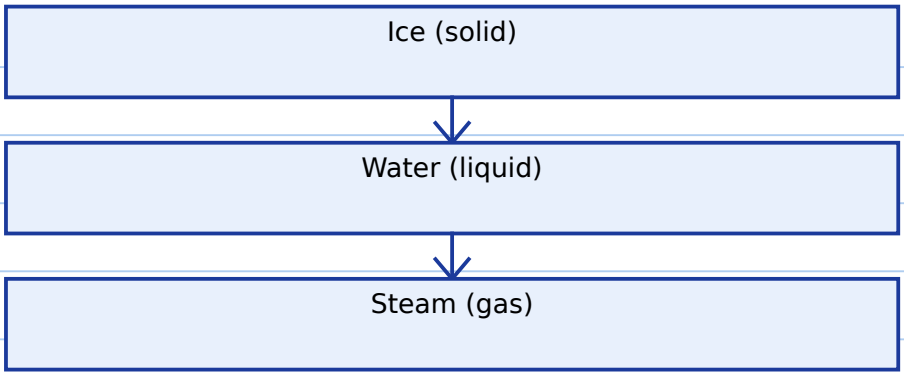
Answer: When we mix sugar in water, it dissolves completely in it and disappears; it is a soluble solid. On the other hand, plastic pieces do not

dissolve in water and, being light, they keep floating on the surface of the water; they are insoluble solids.

Question: 2. (D) Explain with examples: A gas changes into liquid.

Answer: When a gas is cooled, it changes into a liquid; this process is called condensation. Example: drops of water appear on the outer side of a cold glass, because the water vapours present in the air touch the cold surface and change into water. Dew in winter is also formed in the same way.

Matter Changes its State on Heating



Project: Help Saad and Fatima in Separating the Salt from Sand and Tea Leaves

Question: Project: In the kitchen, salt and tea leaves got mixed with sand. How can you help Saad and Fatima in separating the salt from sand and tea leaves from the sand and salt?

Answer: First sieve the mixture; the large tea leaves will stay in the sieve and the salt and sand will pass down. Then mix the salt and sand mixture in water; the salt will dissolve in the water and the sand will not dissolve. Then filter it through a cloth or filter paper; the sand will stay in the filter and the salty water will come down. At last heat the salty water so that the water evaporates away, and white salt will be left in the dish.

Heat and its Measurement

Review Questions: 1. Write Whether Each of the Following Statements is True or False

✗ 1. (a) The highest temperature in clinical thermometer is 32 °C.

✓ 1. (b) Normal body temperature is 98.4 °F or 37 °C.

✗ 1. (c) Heat tells us how hot an object is.

✓ 1. (d) Constriction in the clinical thermometer helps in preventing the flow of mercury downwards.

✗ 1. (e) Temperature flows from higher region to lower region.

2. Tick the Correct Answer

2.

✓ (a) The properties used for heat:

✓ (a) hot, energy, cold

(b) colour, smell, sound

(c) heavy, light, hard

(d) circle, square, point

2. (b) The lowest temperature in laboratory thermometer is:

(b) The lowest temperature in laboratory thermometer is:

(a) 10 °C

(b) 0 °C

(c) 100 °C

✓ (d) -10 °C

2. (c) Which of the following liquids is used in the clinical thermometer?

(c) Which of the following liquids is used in the clinical thermometer?

(a) Alcohol

(b) Mercury

(c) Water

(d) Oil

2. (d) You should NOT do one of the following while handling a clinical thermometer:

(d) You should NOT do one of the following while handling a clinical thermometer:

(a) Stay at one place

(b) Wash it before use

(c) Run

(d) Place it in cover

3. Compare Heat and Temperature: Similarities and Differences

Question: 3. Compare Heat and Temperature. How are heat and temperature similar and different? Write their similarities and differences.

Answer: Similarities: both are related to each other and both are about an object being hot or cold. Differences: heat is a form of energy which flows from a hot object to a cold object, while temperature tells us how hot or cold an object is and it is measured with a thermometer.

4. For Each Figure, Show the Direction of Heat Flow

Question: 4. (i) The temperature of tea in a cup is 70 °C and the surrounding temperature is 0 °C. In which direction will the heat flow?

Answer: Heat will flow from the tea to the surroundings, because the tea is much hotter than the surroundings. So with time the tea will become cold and its temperature will decrease; arrows will be drawn from the tea outwards.

Question: 4. (ii) The temperature of juice in a glass is 5 °C and the surrounding temperature is 0 °C. In which direction will the heat flow?

Answer: Heat will flow from the juice to the surroundings, because the temperature of the juice, 5 °C, is higher than the temperature of the surroundings, 0 °C. Although the difference is small, heat always flows from higher temperature to lower temperature.

Question: 4. (iii) The temperature of a heater is 200 °C. In which direction will the heat flow?

Answer: Heat will flow from the heater to the cold air of the room and to the surroundings, so the whole room will become warm. Arrows will be drawn from the heater outwards in all directions, because heat flows from a hot object to a cold object.

Question: 4. (iv) The temperature at the Sun's surface is 5,500 degrees Celsius. In which direction will the heat flow?

Answer: Heat will flow from the Sun outwards into space and towards the things around it, including the Earth. The Sun is the hottest of all, so its heat spreads in all directions and gives light and warmth to the Earth.

Activity Questions: About the Beakers and Body Temperature

Question: Activity Questions: Measure the temperature of three beakers with a laboratory thermometer and your and your partner's body temperature with a clinical thermometer. Which beaker has the highest temperature and which has the lowest? Is your body temperature different from your friend's body temperature?

Answer: This is a practical activity; the result depends on the kind of water: the beaker with hot water showed the highest temperature and the beaker with icy cold water showed the lowest temperature. The body temperature of every person can be a little different, but the temperature of healthy people remains about 37 °C.

Project: Make Your Own Thermometer

Question: Project: Using tap water, rubbing alcohol, red food colour, a narrow-necked plastic bottle, a clear plastic drinking straw and modelling clay, how will you make your own thermometer, and how will it work?

Answer: Fill the bottle up to one fourth with equal amounts of water and alcohol mixed together and add two drops of food colour. Put the straw in the bottle so that its lower end does not touch the bottom, and seal the mouth of the bottle tightly with modelling clay, keeping the top end of the straw open. When you warm the bottle with your hands, the mixture will get warm and rise up in the straw, and at a cold place it will come down. In this way it shows us the change of temperature.

Force and Machines

Review Questions: 1. Fill in the Blanks

1. (i) When we move, we do one kind of work or the other.

1. (ii) A pulley is used to raise a flag.

1. (iii) Friction is always in the opposing direction to any ongoing action.

1. (iv) The force that pulls everything towards the Earth is called _____.

Answer: gravitational force. It is because of this force that everything falls down to the Earth and we can stand on the Earth.

1. (v) An object changes its shape when force is applied.

2. Answer the Following

Question: 2. (a) If a train travels a distance of 12 kilometers in 2 hours, what is its speed?

Answer: The distance travelled by the train is 12 kilometres and the time taken is 2 hours. To find the speed, the distance is divided by the time, that is, 12 kilometres divided by 2 hours gives 6. So the speed of the train is 6 kilometres per hour.

Question: 2. (b) Name three machines that help us to cut an object into two pieces.

Answer: Three machines that help us in cutting are: the knife, the scissors and the axe (or saw). All these machines work on the principle of the

wedge, because one edge of them is sharp and the other edge is thick.

Question: 2. (c) Write some uses of a wheelbarrow.

Answer: With a wheelbarrow (a trolley with one wheel) we can carry heavy things from one place to another easily. It is used a lot for carrying soil, plants and tools in the garden and for carrying bricks and sand at a building site. Because of its one wheel and two handles, even a heavy load can be moved with little force.

Question: 2. (d) How is movement affected when a force is applied?

Answer: When force is applied, a standing object can start moving and a moving object can become fast, slow or stop. With force, the direction of movement of an object can also change, just as kicking a football sends it to another side. So force is the cause of every kind of change in movement.

3. Tick Mark the Correct Answer

3.

(a) Force can bring the following changes in an object except one:

(a) it changes speed

✓ (b) it changes mass

(c) it changes direction

(d) it changes shape

3. (b) One of the following is used to raise a flag:

(b) One of the following is used to raise a flag:

(a) Screw

(b) Lever

✓ (c) Pulley

(d) Wedge

3. (c) Which of the following statements is correct?

(c) Which of the following statements is correct?

(a) Simple machines magnify the force you use

(b) Simple machines speed up the work and make the work simple

(c) Simple machines may change the direction of the force

(d) Simple machines can make your force more effective

3. (d) A girl walks 150 metres in 5 seconds; her speed is:

(d) A girl walks 150 metres in 5 seconds; her speed is:

(a) 20 m/sec

(b) 30 m/sec

(c) 40 m/sec

(d) 25 m/sec

4. Write Whether Each of the Following Statements is True or False

4. (a) Machine makes work easier.

4. (b) A wheelbarrow is a simple machine.

4. (c) The direction of an object cannot be changed by force.

4. (d) Speed is defined as the rate of change of distance with time.

4. (e) Force is only a push.

5. Identify the Simple Machines Shown Below

Question: 5. Identify the six simple machines shown in the pictures of the book and write their names.

Answer: According to the order of the pictures, the machines are: (1) Wheel and axle — the wheel of a bicycle or a cart, which makes a load easy to move. (2) Lever — a rod like a crowbar, which helps to lift a heavy object. (3) Pulley — a machine with a rope and a wheel, which helps to lift a load up. (4) Screw — which has a grooved thread and joins things firmly. (5) Wedge — a machine with a sharp edge, which cuts or splits things. (6) The axe is also a kind of wedge, which is used for cutting wood. If the

order of the pictures in your book is different, match each name with the shape of its picture.

6. Compare Simple and Compound Machines

Question: 6. Compare simple and compound machines. How are simple and compound machines similar and different? Write their similarities and differences.

Answer: Similarity: both machines make our work easier and both need force to work. Difference: a simple machine is made of only one simple part, such as the lever, pulley and wedge, while a compound machine is made by joining two or more simple machines, such as the bicycle and the wheelbarrow.

7. Name the Simple Machines Used in a Bicycle

Question: 7. Which simple machines are used in a bicycle? Write their names and work.

Answer: There are three main simple machines in a bicycle: (1) Wheel and axle — both wheels help the bicycle to move easily. (2) Lever — the handle and brake levers help to change the direction and to stop the bicycle, and the pedals also work like levers. (3) Screw — all the parts of the bicycle are joined firmly with screws. By the joining of all these, the bicycle becomes a compound machine.

Inquiry Projects: Machines at Home and School, and Designing a Load Carrying Cart

Question: Inquiry Project 1: Find different kinds of machines from your home and school and make a list of them.

Answer: Machines found at home: scissors, knife, rolling pin, the pulley of a well/bucket, needle and axe. Machines found at school: the flag pulley,

scissors, paper-cutting machine and the hinges of doors (lever). Write the name of each machine and also write in front of it which simple machine it is and in which work it helps.

Question: Inquiry Project 2: Design a cart for carrying a load of 70 kg using a one-foot long wooden board, small wheels and a strong string, and describe your design.

Answer: Design: Fix four small wheels under the strong wooden board with screws, so that the board becomes a trolley. Tie a handle of strong string at the front side of the board so that the cart can be pulled. Place the load in the middle of the board and tie it with the string so that it does not move. Because of the wheels, even a heavy load of 70 kg will move easily with little force, because the wheels reduce friction.

Sound

Review Questions: 1. Match the Columns

1.

Column-1	Column-2
1. Sound / Vibrations / Pitch / Medium / Noise / Decibel	is the measurement unit of sound
	something through which sound passes
	unwanted sound
	produced by vibrations
	to shake or to move
	the highness or the lowness of a voice

2. Answer the Following Questions

Question: 2. (a) How is sound produced?

Answer: Sound is produced when an object vibrates (shakes). For example, when a drum is beaten, its skin vibrates and sound comes out, and plucking the string of a sitar also makes the string vibrate and produce sound.

Question: 2. (b) When do we experience noise? Give any example.

Answer: When a sound is very loud, unwanted and troubling, we experience it as noise. For example, the horns of vehicles, the noise of a bazaar and the sound of machines are noise, which troubles our ears.

Question: 2. (c) What mediums does sound use to travel?

Answer: Sound travels through three mediums: gas (air), liquid (water) and solids (wood, iron, etc.). Sound travels fastest in solids, then in liquids and slowest in air.

Question: 2. (d) Why can sound not travel in a vacuum?

Answer: Sound needs some medium to travel, because it moves forward by the vibration of particles. In a vacuum there are no particles or air, so sound cannot travel in a vacuum and there is complete silence there.

Question: 2. (e) What are the sources of noise?

Answer: The main sources of noise are: the horns and engines of vehicles, the machines of factories, loud music and loudspeakers, the sound of aeroplanes and the noise of bazaars and fairs. All these make unwanted and loud sounds and increase noise.

Question: 2. (f) Asma lives near an airport where there is noise all day and night. Give her some suggestions to handle the noise.

Answer: Asma should keep the doors and windows of her house closed and hang thick curtains on the windows, because they reduce the sound. While studying, she can use cotton or earplugs in her ears to avoid the noise, and planting trees around the house will also reduce the noise.

Question: 2. (g) Saad listens to music at a high volume using earphones. What effects can listening to music at high volumes for long periods of time have on his health?

Answer: Listening to music at a high volume with earphones for a long time can damage the ears and the hearing ability can become weak. Besides this, problems like headache, dizziness and not being able to concentrate can also occur, so the volume should always be kept low.

Inquiry Project: Make a String Telephone

Question: Inquiry Project: Make a string telephone using two polystyrene cups, two paper clips and a copper string, and tell how it works.

Answer: Make a small hole in the bottom of each cup, pass one end of the string through it and tie it with a paper clip; prepare the second cup in the same way. Then keeping the string straight and tight, one friend puts a cup to his ear and the other speaks into his cup; the sound reaches from one cup to the other by the vibration of the string. This shows that sound can travel through a solid medium (the string).

Question: Inquiry Project (Campaign): How will you run an awareness campaign about noise pollution?

Answer: Put posters and banners about noise pollution in the school and tell the students in the morning assembly that loud sound is harmful for the ears and health. Advise people to blow horns less, listen to music at a low volume and keep silence near hospitals and schools.

Electricity and Magnetism

Review Questions: 1. Write Whether Each of the Following Statements is True or False



1. (a) Plastic wires are used in electrical appliances at home.

1. (b) Circuit is closed when the switch is on.

1. (c) Current is the movement of positive charges.

1. (d) Steel is a non-magnetic material.

1. (e) Like poles repel each other.

2. Make a List of Five Conductors and Five Insulators

Question: 2. Write five conductor and five insulator materials.

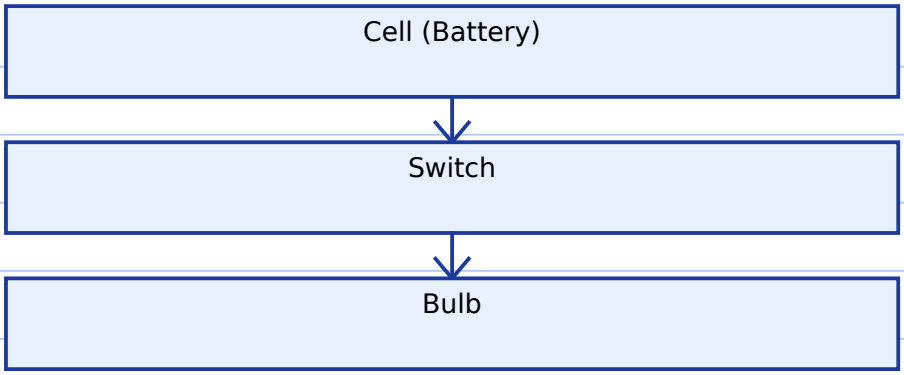
Answer: Conductors are those materials through which electricity passes easily: (1) copper, (2) iron, (3) aluminium, (4) steel and (5) silver. Insulators are those materials through which electricity does not pass: (1) plastic, (2) rubber, (3) wood, (4) glass and (5) paper. That is why wires are made of copper and a layer of plastic or rubber is put over them.

3. Connect a Cell, a Switch and a Bulb to Make a Circuit

Question: 3. How will you connect a cell, a switch and a bulb with wires to make a circuit, and how will it work?

Answer: Take a wire from one end of the cell and join it to the switch, join a wire from the other end of the switch to the bulb and join a wire from the other end of the bulb back to the other end of the cell; in this way a closed round path will be made. When the switch is turned on, the circuit will be closed, electricity will flow and the bulb will light up; turning the switch off will open the circuit and the bulb will go out.

A Simple Electric Circuit



4. Tick Mark the Correct Options

4.

(a) Thing attracted towards magnet:

(a) Copper vessel

(b) Wooden gate

✓ (c) Iron kettle

(d) Copper cup

4. (b) Thing that cannot be used to complete an electric circuit:

(b) Thing that cannot be used to complete an electric circuit:

(a) Copper wire

(b) Iron nail

✓ (c) Wood

(d) Graphite

4. (c) The poles of a suspended magnet point towards _____ respectively:

(c) The poles of a suspended magnet point towards _____ respectively:

✓ (a) North & South of earth

(b) East & West

(c) Up & Down

(d) Right & Left

4. (d) A steel rod can be temporarily magnetise by:

(d) A steel rod can be temporarily magnetise by:

(a) Heating it

(b) Passing electric current around it

(c) Placing it in wax

(d) Hitting it continuously

5. Differentiate Between the Following

Question: 5. (a) Differentiate between open and closed circuits.

Answer: In an open circuit the path of electricity is broken, so current does not flow and the bulb does not light up. In a closed circuit the path is complete and joined, so current flows and the bulb lights up. (Note: in the summary of the book these definitions are printed almost reversed; the scientifically correct definitions are the ones given here.)

Question: 5. (b) Differentiate between permanent and temporary magnets.

Answer: A permanent magnet always keeps its magnetic effect and it cannot be finished, like a bar magnet. A temporary magnet becomes a magnet only when electricity passes through it, and when the electricity is stopped, its effect ends, like an electromagnet.

Question: 5. (c) Differentiate between magnetic and non-magnetic materials.

Answer: Magnetic materials are those which a magnet attracts towards itself, such as iron, steel and nickel. Non-magnetic materials are those which a magnet does not attract, such as wood, plastic, copper and paper.

Question: 5. (d) Differentiate between conductors and insulators.

Answer: Conductors are those materials through which electricity passes easily, such as copper, iron and aluminium; wires are made from them. Insulators are those materials through which electricity does not pass,

such as plastic, rubber and wood; they are used for safety from electricity and for covering wires.

6. Write Three Uses of Magnets

Question: 6. Write three uses of magnets with examples.

Answer: (1) In a compass, the magnetic needle shows the direction, with which travellers and pilots find their way. (2) A magnet is used to pick up and separate pins, nails and other iron things, such as separating iron from garbage. (3) Magnets are used in the doors and fasteners of fridges and bags so that they stay tightly closed. Besides these, magnets are also used in motors and speakers.

Project: Find the Poles of Unmarked Magnets

Question: Project: If you have two unmarked bar magnets, how will you find their poles? And how will you make a magnetic fishing rod game?

Answer: Finding the poles: first take a marked magnet whose poles are known. Bring one end of the unmarked magnet near the north pole of the known magnet; if that end is pushed away (repelled), it is also the north pole, and if it is attracted, it is the south pole. In the same way test the other end too and mark N and S on both. Game: put paper clips on paper fish, tie a string to a rod and fix a small magnet at its end; now play by "catching" the fish with the magnet, because the magnet will attract the clips.

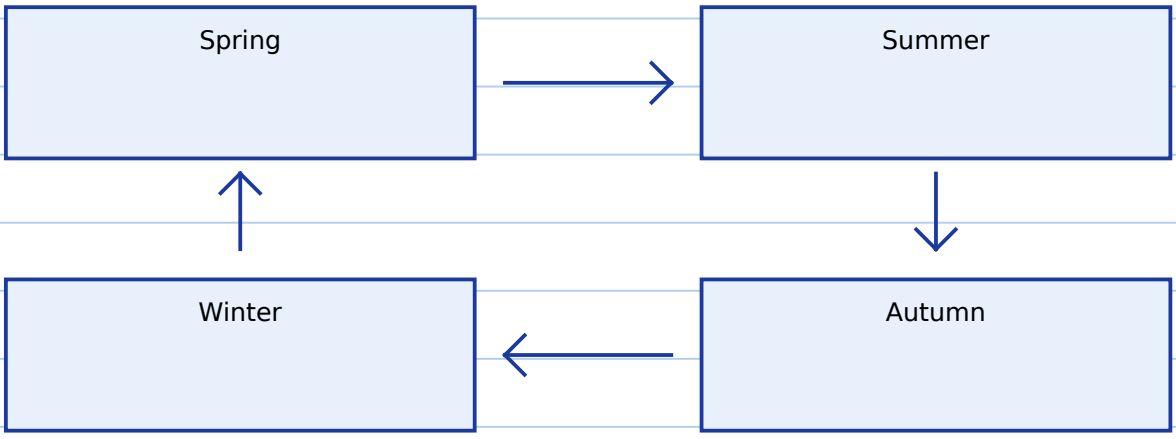
Movements of the Earth

Activity 10.5: Fill in the Blanks (rotation, revolution, Sun, seasons etc.)

Activity 10.5: Fill in the blanks in the following paragraph: The Earth is a _____. It makes two movements i.e. _____ and _____. The Earth revolves around the _____. Due to rotation and revolution _____ are formed. In the part of the Earth tilted towards the Sun there will be _____ and in the part away there will be _____. In the tilted part days will be _____ and nights will be _____, whereas in the other part days will be _____. In _____ and _____ days and nights will be equal on both sides.

Answer: Planet., Rotation, Revolution., Sun., Seasons, Summer, Winter., Hotter, Shorter,, Colder., Spring, Autumn

The Four Seasons of the Year



Review Questions: 1. Circle T for True or F for False for Each Statement and Write the Correct Statements for the False Ones

✗ 1. (i) The Earth is not a sphere.

1. (ii) The Earth has poles, hemispheres and the equator.

1. (iii) The Earth has only one movement which is rotation.

1. (iv) The rotation of Earth takes one year and 6 hours.

1. (v) The revolution of Earth takes 24 hours.

1. (vi) The side of the Earth which is away from the Sun has night.

1. (vii) Due to the revolution of the Earth, seasons and day and night are formed.

1. (viii) When one hemisphere of the Earth is tilted towards the Sun, it is summer there.

1. (ix) On 21st or 22nd March, the day and night are equal in both hemispheres.

1. (x) The Earth has no poles.

2. Word Search: Find the Words and Write Their Meanings

Question: 2. Find the words of the chapter from the word search grid (THE EARTH, AXIS, DAY, EQUATOR, HEMISPHERES, NIGHT, ORBIT, POLES, ROTATION, SEASONS, SPHERE) and write the meaning of each word.

Answer: The words hidden in the grid are: (1) THE EARTH — the planet on which we live. (2) AXIS — the imaginary line on which the Earth rotates. (3) DAY — the side of the Earth where the light of the Sun falls. (4) EQUATOR — the imaginary line in the middle of the Earth. (5) HEMISPHERES — the two halves of the Earth. (6) NIGHT — the side of the Earth where light does not fall. (7) ORBIT — the path of the Earth around the Sun. (8) POLES — the two ends of the Earth, North and South. (9) ROTATION — the spinning of the Earth on its axis. (10) SEASONS — the different periods of the year, such as summer and winter. (11) SPHERE — a completely round shape.

3. Tick the Correct Answer

3. (i) The earth is tilted at?

✓ (a) 23.5°

(b) 50°

(c) 60°

(d) 66°

3. (ii) How long does the Earth take to complete one rotation?

(a) Half an hour

(b) 7 hours

✓ (c) 24 hours

(d) One month

3. (iii) We have shorter days in:

(a) Spring

(b) Summer

(c) Autumn

✓ (d) Winter

3. (iv) We have longer days in:

(a) Winter

(b) Spring

✓ (c) Summer

(d) Autumn

3. (v) The elongated circular orbit of the Earth is called an:

✓ (a) Ellipse

(b) Circle

(c) Square

(d) Straight

3. (vi) The Earth completes one revolution around the sun in:

(a) 28 days

(b) 24 hours

(c) 6 hours

✓ (d) 365 days and 6 hours

3. (vii) An axis is a:

✓ (a) Line

(b) Body

(c) Object

(d) Sphere

3. (viii) The part of the earth that faces the sun will have:

(a) Evening

(b) Night

✓ (c) Day

(d) Nothing

3. (ix) Rotating direction of the Earth is towards the:

(a) South

(b) North

(c) West

✓ (d) East

4. Double Puzzle: Unscramble the Words and Write the Sentence

Question: 4. Unscramble the following letters to make words: TOTIRNOA, SAUSEC, YDA, NAD, HINTG — and then make a sentence from these words.

Answer: The unscrambled words are: TOTIRNOA = ROTATION, SAUSEC = CAUSES, YDA = DAY, NAD = AND, HINTG = NIGHT. The sentence made from them is: ROTATION CAUSES DAY AND NIGHT; it is because of the rotation of the Earth that day and night are formed.

Inquiry Project: Make a Hanging Model of the Solar System

Question: Inquiry Project: Using nine small balls, one large yellow ball, one

cardboard circle piece, a long string and scissors, how will you make a hanging model of the solar system?

Answer: Make the large yellow ball the Sun and colour the eight small balls as the eight planets, writing the name of each on it; make the ring of Saturn around it from the cardboard circle piece. Tie each ball with strings of different lengths to the big cardboard piece, hanging the Sun in the middle and the planets around it in order. The planets in order are: Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus and Neptune. Hang the model from the roof of the class and also fix a name card for each planet.